Between Sector Misallocation

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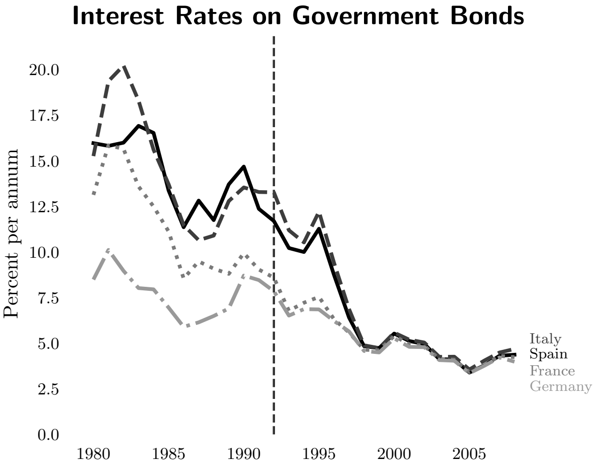
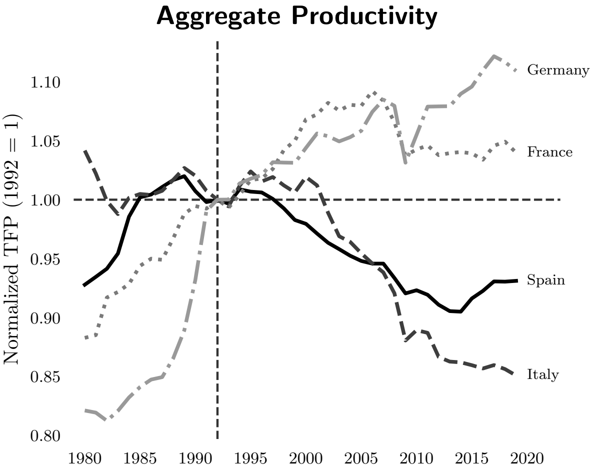
Introduction

- The allocation of economic activity across sectors is central to aggregate productivity dynamics.
- Sectoral composition often changes during periods of structural transformation, trade liberalization, and credit booms.
- Whether these reallocations raise or lower productivity depends on the structural differences and frictions that shape how resources move across sectors.

Falling interest rates and declining productivity

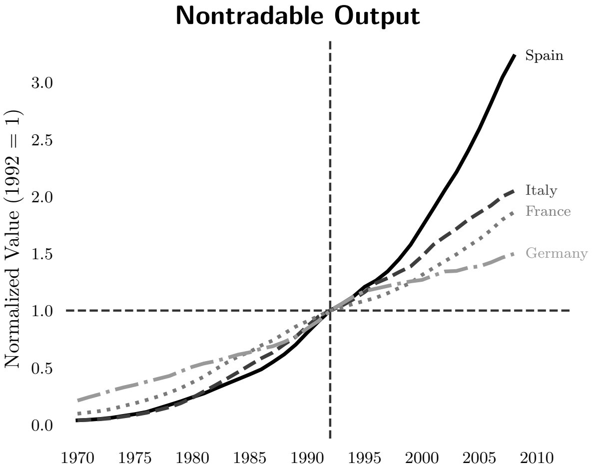


Source: IMF International Financial Statistics

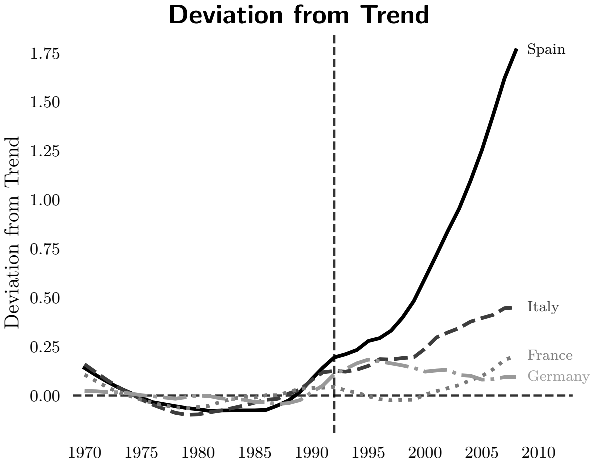


Source: Penn World Table

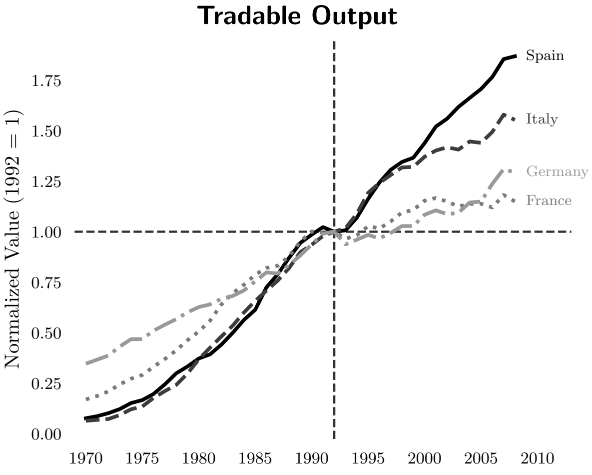
Nontradables expand during the capital inflow episode



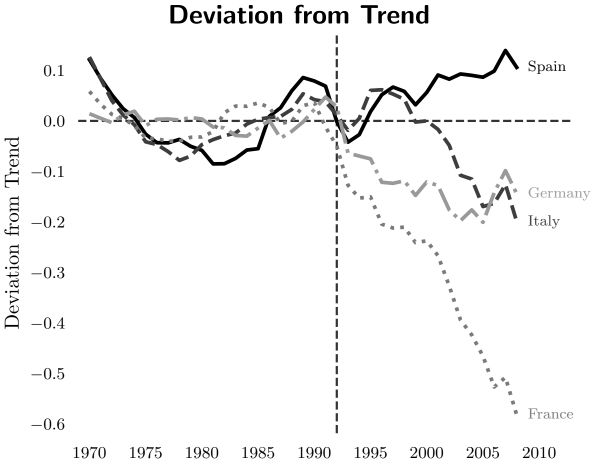
Source: Historical EU KLEMS



Tradables do not expand commensurately



Source: Historical EU KLEMS



This Paper

How do sectoral differences determine the allocation and productivity effects of capital inflows?

- Competing explanations:
    - Low competition in nontradables distorts allocation
      (Brunnermeier & Reis, 2023; Dias et al., 2016, 2020; Reis, 2013)
    - Capital inflows crowd out high-productivity tradable investment
      (Benigno & Fornaro, 2014; Benigno et al., 2025)
    - Tighter borrowing limits in nontradables
      (Ozhan, 2020; Schneider & Tornell, 2004; Tornell & Westermann, 2002)
- Understanding these mechanisms is crucial for **policy**

This Paper

- Develop a **multi-sector quantitative model** with rich firm- and sector-level heterogeneity:
  - Sectoral differences in technology, competition, risk
  - Borrowing constraints, capital adjustment costs, sectoral mobility costs
- Let the data discipline the importance of each mechanism by comparing model-simulated and observed dynamics
- I find that **capital frictions** are key to explaining sectoral reallocation and declining allocative efficiency, while competition and credit frictions play a more modest role

Conceptual Framework

- Builds on Hsieh–Klenow: heterogeneous firms produce differentiated varieties under monopolistic competition.

- This paper adds **sectoral heterogeneity** and **sector-wide distortions** to study misallocation across sectors.

Across sectors

$$Y_t = \left(\sum_{s=1}^{S} Y_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}$$

Within sectors

$$Y_{st} = \left(\sum_{i=1}^{N_{st}} y_{ist}^{\frac{\theta_s-1}{\theta_s}} \right)^{\frac{\theta_s}{\theta_s-1}}$$

- Firm-level demand:

$$y_{ist} = \left(\frac{p_{ist}}{P_{st}} \right)^{-\theta_s} \left(\frac{P_{st}}{P_t} \right)^{-\eta} Y_t$$

CES Aggregation Why CES Firm Demand

Firm and Sector Wedges

- **Sector-wide wedges** create systematic differences in marginal products **across** sectors, while **firm-level wedges** create dispersion **within** sectors.

- Firm i in sector s chooses price, capital, and labor:

$$\max_{p,k,l} (1 - \tilde{\tau}^y_{ist}) p_{ist} y_{ist} - (1 + \tilde{\tau}^k_{ist})(r_t + \delta_s) k_{ist} - w_{st} l_{ist}$$

- Composite wedges combine firm-specific and sector-wide distortions:

$$1 - \tilde{\tau}^y_{ist} = (1 - \tau^y_{ist})(1 - \tau^y_{st}) \qquad 1 + \tilde{\tau}^k_{ist} = (1 + \tau^k_{ist})(1 + \tau^k_{st})$$

FOCs TFPR

Aggregation and Productivity Losses

- **Aggregate productivity** falls when distortions lower sectoral TFP or shift activity toward sectors with lower productivity.

- Sectoral TFP depends on physical productivity and distorted firm scale:

$$TFP_{st} = \left(\sum_i \left(z_{ist} \frac{TFPR_{st}}{TFPR_{ist}} \right)^{\theta_s - 1} \right)^{\frac{1}{\theta_s - 1}}$$

- Aggregate TFP combines sectoral TFP and sectoral composition:

$$TFP_t = \left(\sum_{s=1}^{S} \omega_{st} \, TFP_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}$$

Sectoral TFP TFP Loss Aggregate TFP

Illustrative Model

- Simple model to build intuition of how wedges affect both **input choice** and **sectoral sorting**
- Static economy with heterogeneous firms and sectors:

$$y = zk \qquad y = p^{-\theta_s} \qquad \mu_s = \frac{\theta_s}{\theta_s - 1}$$

- Firms first choose a sector, then choose capital before the realized capital wedge is observed:

$$s^\star = \arg\max_s \mathbb{E}[\pi(s)]$$

- Conditional on sector s, capital solves:

$$\pi(s) = \max_k \mathbb{E}\left[(zk)^{\frac{\theta_s - 1}{\theta_s}} - (1 + \tilde{\tau}^k) rk\right]$$

Wedges Cases

Effect on Input Choice

- Capital wedges reduce capital relative to the efficient benchmark:

$$\frac{k_s^*}{k_s^e} = \left(\frac{1}{\mathbb{E}(1+\tilde{\tau}_s^k)}\right)^{\theta_s}$$

- The loss is larger in high-elasticity sectors:

$$\frac{\partial(k_s^*/k_s^e)}{\partial\theta_s} < 0 \qquad \text{when} \qquad \mathbb{E}(1+\tilde{\tau}_s^k) > 1$$

- Since

$$\mathbb{E}(1+\tilde{\tau}_s^k) = 1 + \mathbb{E}(\tau_i^k) + \mathbb{E}(\tau_s^k) + \mathbb{E}(\tau_i^k)\mathbb{E}(\tau_s^k) + \mathrm{Cov}(\tau_i^k, \tau_s^k)$$

positive covariance between firm- and sector-level wedges *amplifies* the capital shortfall.

Derive k^* | Derive k^*/k^e

Effect on Firm Sorting

- Consider two sectors, A and B, with $\theta_A > \theta_B > 1$. A firm chooses the high-elasticity sector A if $z > \bar{z}$.

$$\bar{z} = \underbrace{r}_{\text{cost of capital}} \cdot \underbrace{\left(\frac{\theta_B - 1}{\theta_A - 1} \cdot \frac{\mu_A^{\theta_A}}{\mu_B^{\theta_B}} \right)^{\frac{1}{\theta_A - \theta_B}}}_{\text{scale gains vs. markups}} \cdot \underbrace{\frac{\left[\mathbb{E}(1+\tilde{\tau}_A^k)\right]^{\frac{\theta_A - 1}{\theta_A - \theta_B}}}{\left[\mathbb{E}(1+\tilde{\tau}_B^k)\right]^{\frac{\theta_B - 1}{\theta_A - \theta_B}}}}_{\text{relative capital distortions}}$$

- High-productivity firms sort into high-elasticity sectors because profits scale as $z^{\theta_s - 1}$.
- A higher capital wedge in sector A raises $\bar{z}$ and pushes marginal firms away from the high-elasticity sector.
- Sorting is distorted when the sector that offers the largest scale gains is also the sector with tighter capital frictions.

Profits Threshold

Effect of Capital Flows

- Model capital inflows as a decline in the cost of capital: $r \downarrow$
- **Intensive margin:** capital expands more in high-elasticity sectors:

$$\frac{\partial \ln k_s^*}{\partial \ln r} = -\theta_s$$

- **Extensive margin:** lower r reduces the productivity threshold for the high-elasticity sector:

$$\bar{z} \propto r \qquad \Longrightarrow \qquad \frac{\partial \ln \bar{z}}{\partial \ln r} = 1$$

- Capital inflows therefore operate through two allocation margins:
  - within-sector capital deepening;
  - between-sector re-sorting of marginal firms.
- Relative sectoral wedges determine whether the reallocation improves or worsens allocative efficiency.

Quantitative Model

- **Objective:** use the quantitative model to evaluate competing channels behind the observed dynamics and quantify their relative importance.
- The model combines heterogeneous firms with sectoral differences in technology, market power, risk, and financial frictions.
- Input wedges arise endogenously from firms' dynamic responses to the constraints and distortions they face.
- Capital inflows modeled as exogenous decline in the real interest rate.

Primitives

Recursive Formulation

- Entrepreneurs differ in sector, net worth, capital, and productivity:

$$\mathbf{x} = (s, a, k) \qquad \mathbf{z} = (z^P, z^T_T, z^T_N)$$

- Given aggregate prices $\boldsymbol{\Gamma} = (w, r)$, the entrepreneur solves:

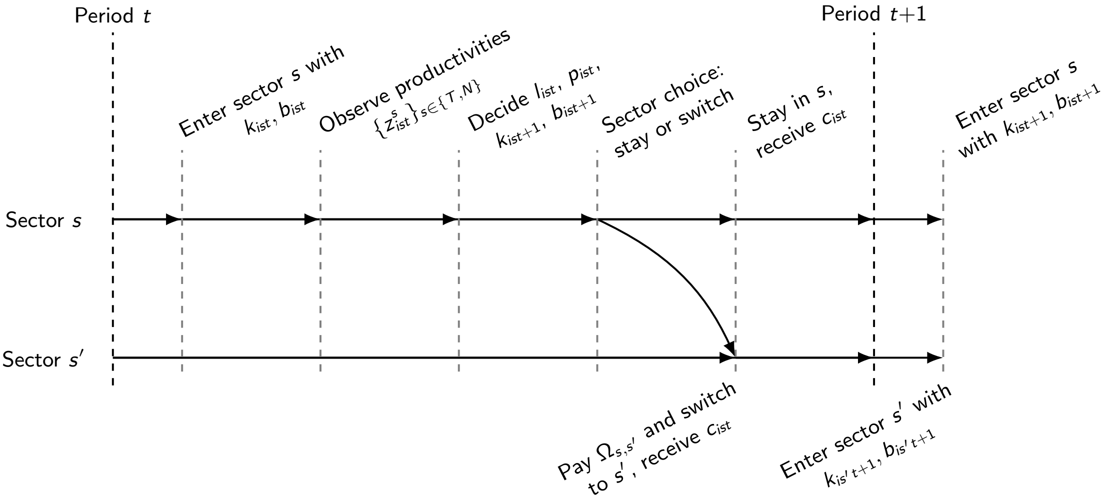
$$V(\mathbf{x}, \mathbf{z}; \boldsymbol{\Gamma}) = \max_{s', c, a', k', l, p} \{u(c) + \beta \mathbb{E}_{\mathbf{z}'} V(\mathbf{x}', \mathbf{z}'; \boldsymbol{\Gamma}')\}$$

$$c + a' + \underbrace{\Phi_s(k', k)}_{\text{capital adjustment cost}} + \underbrace{\Omega_{s,s'}(k)}_{\text{sector mobility cost}} = p_s y_s - wl - (r + \delta)k - \kappa_s + (1 + r)a$$

$$y_s = z_s k^{\alpha_s} l^{1-\alpha_s} \qquad y_s = p_s^{-\theta_s} \qquad k' \leq \lambda_s a'$$

Technology Productivity Frictions Budget Labor

Timing Protocol



Transition Experiment

- **Objective:** calibrate the model to match Spain's post-Maastricht transition.

- For each candidate parameter vector Θ:

  1. solve the stationary equilibrium under the 1992 real interest rate;
  2. simulate the transition using the observed path of real interest rates;
  3. compare model-generated paths with empirical paths.

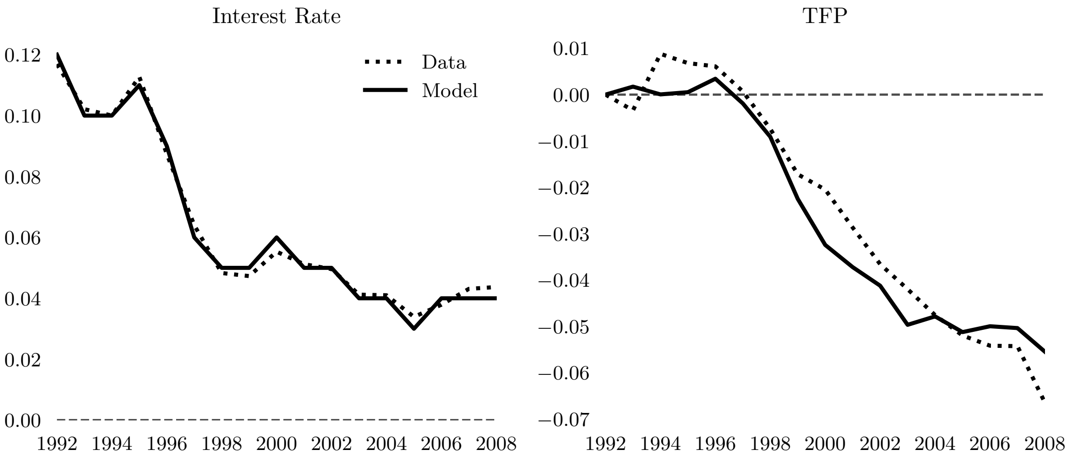
- Target moments are log changes in TFP_t, Y_t^T, Y_t^N

- The calibrated parameters minimize the distance between model and data:

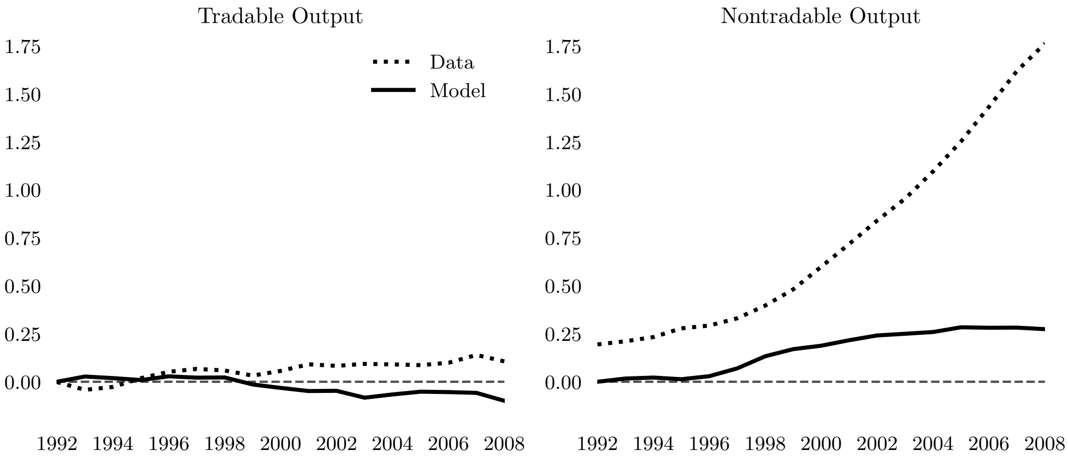
$$\Theta^* = \arg\min_{\Theta} M(\Theta)' W M(\Theta)$$

Equilibrium SMM

Quantitative Fit



Quantitative Fit



Quantitative Fit

- Modest sectoral differences in **competition** and **credit constraints**:
  - Demand elasticity (θ) 10% higher in tradables
  - Borrowing constraint (λ) 9% more relaxed in tradables
- Larger asymmetries in **capital frictions**:
  - Capital adjustment costs (ϕ) 53% higher in tradables
  - Sector switching costs (ω) 55% higher in tradables
- Capital frictions dampen tradable sector expansion in response to falling r:
  - Capital adjustment costs limit *intensive* margin
  - Mobility costs limit *extensive* margin

▸ Fit ▸ Parameters

Conclusion

- Capital inflows can reduce productivity by shifting activity toward nontradables, but the underlying mechanisms remain unclear
- I develop a multi-sector model with rich heterogeneity to quantify sources of sectoral reallocation and TFP decline
- I find that **capital frictions** play a central role in explaining the observed dynamics
- From a **policy** perspective, facilitating capital mobility can raise productivity by improving reallocation of inputs

Extra Slides

CES Aggregation

Final good aggregation

$$Y_t = \left(\sum_{s=1}^{S} Y_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} \qquad \eta > 1$$

Aggregate price index

$$P_t = \left(\sum_{s=1}^{S} P_{st}^{1-\eta} \right)^{\frac{1}{1-\eta}}$$

Sectoral expenditure shares

$$\frac{P_{st} Y_{st}}{P_t Y_t} = \left(\frac{P_{st}}{P_t} \right)^{1-\eta}$$

Back

Why CES Across Sectors?

Cobb–Douglas across sectors

$$Y_t = \prod_s Y_{st}^{\eta_s} \qquad P_{st} Y_{st} = \eta_s P_t Y_t$$

- Expenditure shares fixed.
- Sectoral productivity changes absorbed by prices.
- Input allocation across sectors does not respond through expenditure shares.

CES across sectors

$$Y_t = \left(\sum_s Y_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} \qquad P_t = \left(\sum_s P_{st}^{1-\eta} \right)^{\frac{1}{1-\eta}}$$

- Expenditure shares respond to relative prices.
- Sectoral productivity affects sectoral size.
- Departure necessary to study intersectoral allocation.

Back

Firm Demand

Within-sector aggregation

$$Y_{st} = \left(\sum_{i=1}^{N_{st}} y_{ist}^{\frac{\theta_s - 1}{\theta_s}} \right)^{\frac{\theta_s}{\theta_s - 1}} \qquad \theta_s > 1$$

Sectoral price index

$$P_{st} = \left(\sum_{i=1}^{N_{st}} p_{ist}^{1-\theta_s} \right)^{\frac{1}{1-\theta_s}}$$

Firm-level demand

$$y_{ist} = \left(\frac{p_{ist}}{P_{st}} \right)^{-\theta_s} \left(\frac{P_{st}}{P_t} \right)^{-\eta} Y_t$$

Back

Firm First-Order Conditions

Firm problem

$$\max_{p_{ist}, k_{ist}, l_{ist}} \quad (1 - \tilde{\tau}^y_{ist}) p_{ist} y_{ist} - (1 + \tilde{\tau}^k_{ist})(r_t + \delta_s) k_{ist} - w_{st} l_{ist}$$

Production

$$y_{ist} = z_{ist} k_{ist}^{\alpha_s} l_{ist}^{1-\alpha_s}$$

FOC

$$MRPL_{ist} = \left(\frac{1-\alpha_s}{\mu_s}\right) \frac{p_{ist} y_{ist}}{l_{ist}} = \frac{1}{1-\tilde{\tau}^y_{ist}} w_{st}$$

$$MRPK_{ist} = \left(\frac{\alpha_s}{\mu_s}\right) \frac{p_{ist} y_{ist}}{k_{ist}} = \frac{1+\tilde{\tau}^k_{ist}}{1-\tilde{\tau}^y_{ist}} (r_t + \delta_s)$$

Back

TFPR and Marginal Products

Revenue productivity

$$TFPR_{ist} = p_{ist} z_{ist} = \frac{p_{ist} y_{ist}}{k_{ist}^{\alpha_s} l_{ist}^{1-\alpha_s}}$$

Equivalent expression using marginal products

$$TFPR_{ist} = \mu_s \left(\frac{MRPK_{ist}}{\alpha_s} \right)^{\alpha_s} \left(\frac{MRPL_{ist}}{1-\alpha_s} \right)^{1-\alpha_s}$$

Substituting the FOCs

$$TFPR_{ist} = \mu_s \left[\frac{1+\tilde{\tau}_{ist}^k}{1-\tilde{\tau}_{ist}^y} \frac{r_t + \delta_s}{\alpha_s} \right]^{\alpha_s} \left[\frac{1}{1-\tilde{\tau}_{ist}^y} \frac{w_{st}}{1-\alpha_s} \right]^{1-\alpha_s}$$

Back

Sectoral TFP

Sectoral inputs

$$K_{st} = \sum_{i=1}^{N_{st}} k_{ist} \qquad L_{st} = \sum_{i=1}^{N_{st}} l_{ist}$$

Sectoral TFP

$$TFP_{st} = \frac{Y_{st}}{K_{st}^{\alpha_s} L_{st}^{1-\alpha_s}} = \left(\sum_{i=1}^{N_{st}} \left(z_{ist} \frac{TFPR_{st}}{TFPR_{ist}} \right)^{\theta_s - 1} \right)^{\frac{1}{\theta_s - 1}}$$

Efficient benchmark

$$TFPR_{ist} = TFPR_{st} \quad \forall i \qquad \Rightarrow \qquad TFP_{st}^{e} = \left(\sum_{i=1}^{N_{st}} z_{ist}^{\theta_s - 1} \right)^{\frac{1}{\theta_s - 1}}$$

▸ Back

TFP Loss from Misallocation

Distorted sectoral TFP

$$TFP_{st} = \left(\sum_i \left(z_{ist} \frac{TFPR_{st}}{TFPR_{ist}} \right)^{\theta_s - 1} \right)^{\frac{1}{\theta_s - 1}}$$

Efficient sectoral TFP

$$TFP^e_{st} = \left(\sum_i z_{ist}^{\theta_s - 1} \right)^{\frac{1}{\theta_s - 1}}$$

Log TFP loss

$$\log \left(\frac{TFP_{st}}{TFP^e_{st}} \right) = \frac{1}{\theta_s - 1} \left[\log \left(\sum_i z_{ist}^{\theta_s - 1} \left(\frac{TFPR_{st}}{TFPR_{ist}} \right)^{\theta_s - 1} \right) - \log \left(\sum_i z_{ist}^{\theta_s - 1} \right) \right]$$

Back

Aggregate TFP

Aggregate inputs

$$K_t = \sum_{s=1}^{S} K_{st} \qquad L_t = \sum_{s=1}^{S} L_{st}$$

Aggregate output

$$Y_t = \left(\sum_{s=1}^{S} Y_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} = \left(\sum_{s=1}^{S} \left[TFP_{st} K_{st}^{\alpha_s} L_{st}^{1-\alpha_s} \right]^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}$$

Aggregate TFP

$$TFP_t = \left(\sum_{s=1}^{S} \omega_{st} TFP_{st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} \qquad \omega_{st} = \frac{Y_{st}^{\frac{\eta-1}{\eta}}}{\sum_{s'=1}^{S} Y_{s't}^{\frac{\eta-1}{\eta}}}$$

Back

Capital Wedges

Composite capital wedge

$$1+\tilde{\tau}_s^k = (1+\tau_i^k)(1+\tau_s^k)$$

Expected capital wedge

$$\begin{aligned}\mathbb{E}(1+\tilde{\tau}_s^k) &= \mathbb{E}\left[(1+\tau_i^k)(1+\tau_s^k)\right] \\ &= 1+\mathbb{E}(\tau_i^k)+\mathbb{E}(\tau_s^k)+\mathbb{E}(\tau_i^k\tau_s^k)\end{aligned}$$

Using covariance

$$\mathbb{E}(1+\tilde{\tau}_s^k) = 1+\mathbb{E}(\tau_i^k)+\mathbb{E}(\tau_s^k)+\mathbb{E}(\tau_i^k)\mathbb{E}(\tau_s^k)+\mathrm{Cov}(\tau_i^k,\tau_s^k)$$

Back

Correlation Between Wedges

Composite capital wedge

$$1 + \tilde{\tau}_s^k = (1 + \tau_i^k)(1 + \tau_s^k)$$

Expected wedge

$$\mathbb{E}(1 + \tilde{\tau}_s^k) = 1 + \mathbb{E}(\tau_i^k) + \mathbb{E}(\tau_s^k) + \mathbb{E}(\tau_i^k)\mathbb{E}(\tau_s^k) + \mathrm{Cov}(\tau_i^k, \tau_s^k)$$

Perfect positive correlation

$$\tau_s^k = \tau_i^k \qquad \Rightarrow \qquad \mathbb{E}(1 + \tilde{\tau}_s^k) = 1 + 2\mathbb{E}(\tau_i^k) + \mathbb{E}(\tau_i^k)^2 + \mathrm{Var}(\tau_i^k)$$

Perfect negative correlation

$$\tau_s^k = -\tau_i^k \qquad \Rightarrow \qquad \mathbb{E}(1 + \tilde{\tau}_s^k) = 1 - \mathbb{E}\left[(\tau_i^k)^2\right]$$

Back

Optimal Capital Choice

Firm problem

$$\max_{k} \ \mathbb{E}\left[(zk)^{\frac{\theta_s-1}{\theta_s}} - (1+\tilde{\tau}_s^k)rk\right]$$

FOC

$$\frac{\theta_s-1}{\theta_s} z^{\frac{\theta_s-1}{\theta_s}} k^{-\frac{1}{\theta_s}} = r\mathbb{E}(1+\tilde{\tau}_s^k)$$

Optimal capital

$$k_s^* = \left[\frac{\frac{\theta_s-1}{\theta_s} z^{\frac{\theta_s-1}{\theta_s}}}{r\mathbb{E}(1+\tilde{\tau}_s^k)}\right]^{\theta_s} = \frac{z^{\theta_s-1}}{\mu_s^{\theta_s} r^{\theta_s}\left[\mathbb{E}(1+\tilde{\tau}_s^k)\right]^{\theta_s}}$$

with $\mu_s = \theta_s/(\theta_s-1)$

Back

Capital Gap

Efficient capital choice

$$k_s^e = \frac{z^{\theta_s - 1}}{\mu_s^{\theta_s} r^{\theta_s}}$$

Capital relative to efficient benchmark

$$\frac{k_s^*}{k_s^e} = \left(\frac{1}{\mathbb{E}(1+\tilde{\tau}_s^k)}\right)^{\theta_s} = \left[\frac{1}{1+\mathbb{E}(\tau_i^k)+\mathbb{E}(\tau_s^k)+\mathbb{E}(\tau_i^k)\mathbb{E}(\tau_s^k)+\mathrm{Cov}(\tau_i^k,\tau_s^k)}\right]^{\theta_s}$$

Effect of sectoral demand elasticity

$$\frac{\partial(k_s^*/k_s^e)}{\partial\theta_s} = -\left[\mathbb{E}(1+\tilde{\tau}_s^k)\right]^{-\theta_s}\log\left[\mathbb{E}(1+\tilde{\tau}_s^k)\right] < 0$$

Back

Optimal Profits

Profit at optimal capital

$$\pi^*(s) = \max_k \ \mathbb{E}\left[(zk)^{\frac{\theta_s - 1}{\theta_s}} - (1 + \tilde{\tau}_s^k) rk\right]$$

Using the optimal capital choice

$$k_s^* = \frac{z^{\theta_s - 1}}{\mu_s^{\theta_s} r^{\theta_s} \left[\mathbb{E}(1 + \tilde{\tau}_s^k)\right]^{\theta_s}}$$

Expected profit

$$\pi^*(s) = \frac{(\theta_s - 1)^{\theta_s - 1} z^{\theta_s - 1}}{\theta_s^{\theta_s} r^{\theta_s - 1} \left[\mathbb{E}(1 + \tilde{\tau}_s^k)\right]^{\theta_s - 1}}$$

Productivity elasticity of profits

$$\frac{\partial \ln \pi^*(s)}{\partial \ln z} = \theta_s - 1$$

Back

Sorting Threshold

Sector comparison

$$\theta_A > \theta_B > 1 \qquad \text{and} \qquad \pi^*(A) > \pi^*(B)$$

Expected profits

$$\pi^*(s) = \frac{(\theta_s - 1)^{\theta_s - 1} z^{\theta_s - 1}}{\theta_s^{\theta_s} r^{\theta_s - 1} \left[\mathbb{E}(1 + \tilde{\tau}_s^k)\right]^{\theta_s - 1}}$$

Threshold productivity

$$\bar{z} = r \left(\frac{\theta_B - 1}{\theta_A - 1} \frac{\mu_A^{\theta_A}}{\mu_B^{\theta_B}} \right)^{\frac{1}{\theta_A - \theta_B}} \frac{\left[\mathbb{E}(1 + \tilde{\tau}_A^k)\right]^{\frac{\theta_A - 1}{\theta_A - \theta_B}}}{\left[\mathbb{E}(1 + \tilde{\tau}_B^k)\right]^{\frac{\theta_B - 1}{\theta_A - \theta_B}}}$$

Sorting rule

$$z > \bar{z} \qquad \Rightarrow \qquad s^\star = A$$

Back

Capital Inflows

Capital inflow shock: $r \downarrow$

Intensive margin

$$k_s^* = \frac{z^{\theta_s - 1}}{\mu_s^{\theta_s} r^{\theta_s} \left[\mathbb{E}(1+\tilde{\tau}_s^k)\right]^{\theta_s}} \qquad \Rightarrow \qquad \frac{\partial \ln k_s^*}{\partial \ln r} = -\theta_s$$

Extensive margin

$$\bar{z} \propto r \qquad \Rightarrow \qquad \frac{\partial \ln \bar{z}}{\partial \ln r} = 1$$

$r \downarrow \quad \Rightarrow \quad$ capital expands within sectors and the sorting threshold falls.

Back

Model Primitives

Environment

- Small open economy populated by a large number of monopolistically competitive firms
- Firms hire labor competitively at wage w_t
- Firms save and borrow internationally through a one-period bond traded at interest rate r_t
- No aggregate uncertainty

Entrepreneur preferences

$$\mathbb{E} \sum_{t=0}^{\infty} \beta^t \left(\frac{c_{ist}^{1-\gamma} - 1}{1-\gamma} \right)$$

Back

Technology and Demand

Production

$$y_s = z_s k^{\alpha_s} l^{1-\alpha_s}$$

Demand

$$y_s = p_s^{-\theta_s} \quad \Longleftrightarrow \quad p_s = y_s^{-\frac{1}{\theta_s}}$$

Markup

$$\mu_s = \frac{\theta_s}{\theta_s - 1}$$

Operating profits

$$\pi_s = p_s y_s - wl - \kappa_s$$

Back

Productivity Process

Firm productivity

$$z_{ist} = z_i^P \exp(z_{ist}^T)$$

Permanent productivity

$$F(z_i^P) = 1 - \left(\frac{1}{z_i^P}\right)^{\varphi} \qquad z_i^P \geq 1$$

Sector-specific transitory productivity

$$z_{ist}^T = \rho_s z_{ist-1}^T + \varepsilon_{ist}^T \qquad \varepsilon_{ist}^T \sim N(0, \sigma_s^T)$$

Back

Sectoral Frictions

Borrowing constraint

$$b_{ist} \leq \chi_s k_{ist} \qquad \Longleftrightarrow \qquad k' \leq \lambda_s a' \qquad \lambda_s = \frac{1}{1-\chi_s}$$

Capital adjustment cost

$$\Phi_s(k', k) = \frac{\phi_s}{2}\left(\frac{k' - (1-\delta)k}{k}\right)^2 k$$

Sectoral mobility cost

$$\Omega_{s,s'}(k) = \mathbb{I}_{s' \neq s}\omega_s k$$

Back

Optimal Labor Demand

Static operating profits

$$\pi_s = \left(z_s k^{\alpha_s} l^{1-\alpha_s}\right)^{\frac{\theta_s - 1}{\theta_s}} - wl - \kappa_s$$

Labor FOC

$$\frac{\theta_s - 1}{\theta_s}(1-\alpha_s) z_s^{\frac{\theta_s - 1}{\theta_s}} k^{\frac{\alpha_s(\theta_s - 1)}{\theta_s}} l^{-\frac{1+\alpha_s(\theta_s - 1)}{\theta_s}} = w$$

Optimal labor demand

$$l_s(k, z_s, w) = \left(\frac{(1-\alpha_s)(\theta_s - 1)}{w\theta_s}\right)^{\frac{\theta_s}{1+\alpha_s(\theta_s - 1)}} z_s^{\frac{\theta_s - 1}{1+\alpha_s(\theta_s - 1)}} k^{\frac{\alpha_s(\theta_s - 1)}{1+\alpha_s(\theta_s - 1)}}$$

Back

Budget Constraint

Original budget constraint

$$c + k' - (1-\delta)k + (1+r)b + \Phi_s(k', k) + \Omega_{s,s'}(k) = \pi_s + b'$$

Net worth

$$a = k - b \qquad a' = k' - b'$$

Recursive form

$$c + a' + \Phi_s(k', k) + \Omega_{s,s'}(k) = \pi_s - (r+\delta)k + (1+r)a$$

Using operating profits

$$c + a' + \Phi_s(k', k) + \Omega_{s,s'}(k) = p_s y_s - wl - (r+\delta)k - \kappa_s + (1+r)a$$

Back

Stationary Equilibrium

Given prices $\boldsymbol{\Gamma} = (w, r)$, a stationary partial equilibrium satisfies:

1. **Optimality:** The value function solves:

$$V(\mathbf{x}, \mathbf{z}, \boldsymbol{\Gamma}) = \max_{s', c, a', k', l, p} \left\{ u(c) + \beta \mathbb{E}_{\mathbf{z}'} V(\mathbf{x}', \mathbf{z}', \boldsymbol{\Gamma}') \right\}$$

2. **Feasibility:** Policies satisfy the budget constraint, borrowing constraint, and non-negative net worth condition

3. **Consistency:** Policies map current endogenous states into next-period states:

$$\mathbf{x}' = \left(s'(\cdot), a'(\cdot), k'(\cdot) \right)$$

4. **Invariant distribution:** The firm distribution is stationary:

$$\mu'(\mathbf{x}', \mathbf{z}') = \mathcal{T}\left(\mu(\mathbf{x}, \mathbf{z}) \right) \qquad \mu' = \mu$$

Back

Simulated Method of Moments

Candidate parameter vector

$$\Theta = (\theta_s, \alpha_s, \kappa_s, \rho_s, \sigma_s, \lambda_s, \phi_s, \omega_s)_{s \in \{T,N\}}$$

Model and data moments

$$\mathcal{M}_t^j(\Theta) \qquad \mathcal{D}_t^j \qquad j = 1, \ldots, J \qquad t = 1, \ldots, T$$

Moment errors

$$M(\Theta) = \begin{bmatrix} \mathcal{M}_1^1(\Theta) - \mathcal{D}_1^1 \\ \mathcal{M}_2^1(\Theta) - \mathcal{D}_2^1 \\ \vdots \\ \mathcal{M}_T^J(\Theta) - \mathcal{D}_T^J \end{bmatrix}$$

Objective function

$$\Theta^* = \arg\min_{\Theta} M(\Theta)' W M(\Theta) \qquad W = I$$

Back

Model Fit

Fit by target

Target	**RMSE**
TFP_t	0.006
Y_t^T	0.118
Y_t^N	0.717
Total	0.420

Back

Model Parameters

Assigned Parameters

β	γ	δ	η	φ
0.875	1.50	0.06	1.50	30.0

Fitted Parameters

Parameter	Tradable	Nontradable	% Diff.
Demand elasticity θ_s	4.15	3.75	10%
Capital share α_s	0.34	0.29	16%
Fixed cost κ_s	0.11	0.00	200%
Persistence ρ_s	0.53	0.62	16%
Volatility σ_s	0.28	0.36	25%
Borrowing limit λ_s	1.12	1.02	9%
Adjustment cost ϕ_s	4.51	2.62	53%
Mobility cost ω_s	0.14	0.08	55%

Back